



Engineering a Sustainable aquatic/wetland Biodome

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Project design challenge



Our team is designing a model of a freshwater wetland/aquatic ecosystem.



We needed to create a container which could hold the water for our biome.



Our limits were that we had a limited amount of time and that we wanted to create the tank with the acrylic sheets.



Full Project Materials

Acrylic Case Materials & Cost

material	quantity	purpose	cost
1/4" (6mm) acrylic sheet	5	Tank walls	In lab
Loctite Plastics Bonder	2 bottles	Solvent-welds acrylic seams	12.99
Loctite Clear Silicone	1 tube	waterproof seal	8.99

Aluminum Frame Materials & cost

material	quantity	purpose	cost
Aluminum extrusion, 15" stock (cut to 14"/11")	16 pieces total	Rigid outer frame for both tanks	In lab
Corner brackets/fasteners	as needed per frame corner	Joins frame pieces at corners	In lab

Water and Filtration System (Features)

material	quantity	purpose	cost
pumps	2	filtration	In lab
water	3.7 gallons	The environment	nature
gravel/sand	3 bags of gravel and one bag of sand	Filtration and decoration	In lab
minnows	5	Water quality testers	nature

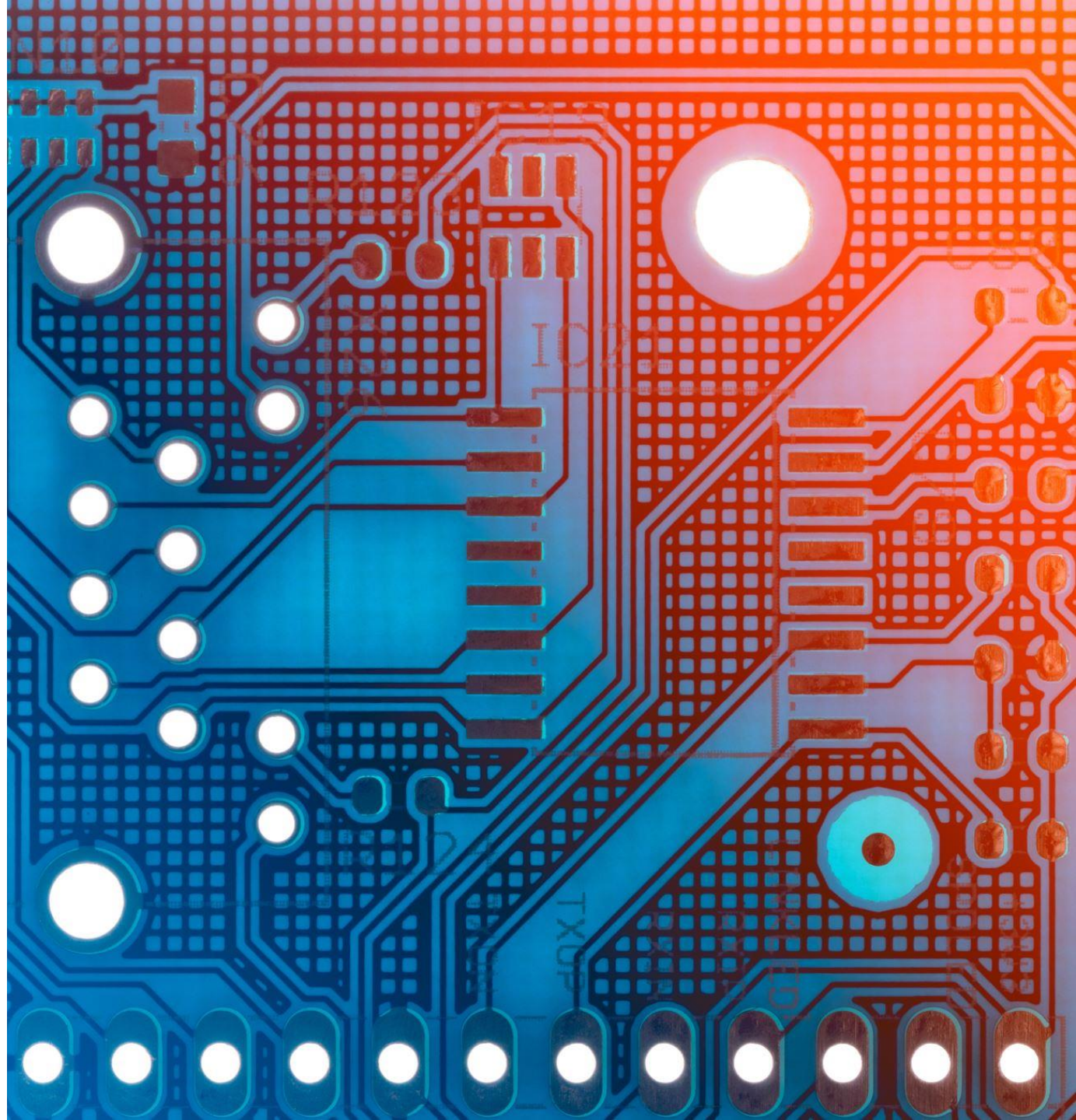
Builds

- We chose this design because it allowed us to fill the container with water without any leaks.
- **Aluminum Frame and Acrylic Case**
- The 15-inch aluminum stock is assembled into a rectangular frame around the outside of each acrylic case using corner brackets/fasteners.
- Each tank's panels are cut to 14" x 11" x 11" dimensions, edges sanded flat, then assembled into a sealed box using acrylic cement. Silicone beads are added afterward on the interior joints for a backup waterproof layer.



Technology Component

- A micro: bit paired with a Hummingbird Bit controller monitors the system.
- A LED-and-light-sensor pair measures the filtration performance.
- A distance sensor tracks the water level from above without touching the water.



Results of our testing

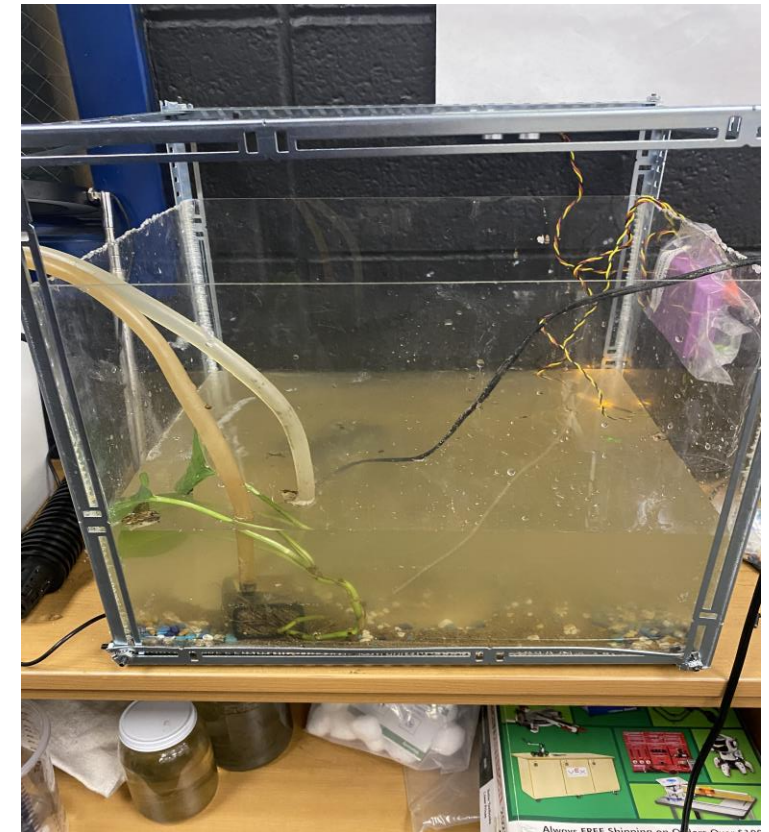
Before

What worked

- The container was able to hold the water without any leaks. The water was able to clear up.

What didn't work

- We originally planned to make two tanks with one being inside the other. The metal frames were too big to allow us to make a second case small enough to fit in the first one.



Reflection

- Some challenges we experienced was working with the microbit. We couldn't get it to display what we wanted at first but it eventually worked.
- We would change the design by using smaller metal frames so that we could've attached another tank to this one. The most important part of the engineering design process was the planning part. It guided us so we knew what to do instead of wasting time trying things without thinking.